



# Preparation and performance properties of polymer latex SDNL in water-based drilling fluids for drilling troublesome shale formations



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## ABSTRACT

In this paper, a novel poly(styrene/n-butyl acrylate/acrylic acid) latex (SDNL) for water-based drilling fluids was successfully prepared and characterized. Emulsion polymerization was used to prepare the polymer latex (SDNL), which was characterized by transmission electron microscopy (TEM), particle size distribution (PSD) and thermal gravimetric analysis (TGA). Firstly, its effect on the rheological and filtration control performance of drilling fluids was analyzed through the compatible test. Secondly, its inhibition performance was observed with the linear swelling test, and the shale cuttings hot-rolling dispersion test. Then, its sealing performance was investigated through the pressure transmission test. Finally, its effect on the lubrication performance was revealed by measuring lubrication coefficients of the drilling fluids.

The results indicated that the particle size distribution of the spherical polymer latex nanoparticles ranged from 50 nm to 190 nm with the D50 value of around 88 nm. SDNL could significantly improve the rheological performance and enhance the shear thinning property of the drilling fluids. SDNL had favorable filtration control performance and good temperature resistance at 120 °C. The inhibition performance test demonstrated that SDNL had excellent ability to inhibit shale hydration swelling and dispersion. At the concentration of 2%, the linear expansion rate of the shale sample was only 4.9%, and the recovery rate was 93.7%. The pressure transmission test revealed that SDNL had outstanding sealing performance. A dense plugging film could form on the shale surface, the permeability of which was almost zero, and the pore pressure was difficult to continually transmit. Furthermore, the lubrication performance results confirmed that SDNL had good lubrication performance. After adding 2.0 w/v% of SDNL into the drilling fluids, the lubrication coefficient was reduced as high as 64.4%. The newly developed polymer latex SDNL is expected to be a superior additive in water-based drilling fluids for drilling troublesome shale formations.

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## 1. Introduction

Wellbore instability is often encountered during oil and gas drilling process, which results in higher drilling costs and delays in the drilling process (Hale et al., 1993; Bol et al., 2013; Van Oort, 1994). Approximately 75% of the formations in the drilling process are shale formations, and 90% of the borehole instability occurs in shale formations (Steiger and Leung, 1992). Thus, borehole instability in shale formations has attracted much attention (Chenevert and Sharma, 1993; Ma and Chen, 2015). Shale is mainly composed of clay minerals and suffers from hydration effect when

drilled with traditional water-based drilling fluids. This leads to a change in the stress state of the borehole wall and a decrease of the rock strength, which is not conducive to wellbore stability. In particular, as shale has extremely low permeability, the pressure transmission and filtrate invasion of drilling fluids will cause an increase in pore pressure and a reduction in the effective support of the liquid column pressure, thus resulting in wellbore instability (Van Oort, 1997; Stowe et al., 2001). Therefore, the inhibition and sealing performance of the traditional water-based drilling fluids should be enhanced (Qiu et al., 2007).

Over the years, inorganic salt, silicate, formate, polyol and amine polymer have been introduced into drilling fluids as inhibitors, while asphalt, silicate and aluminum salts have been used as plugging agents (Caenn and Chillingar, 1996; Ramirez et al., 2007). The inhibition and sealing performance of the traditional water-

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